

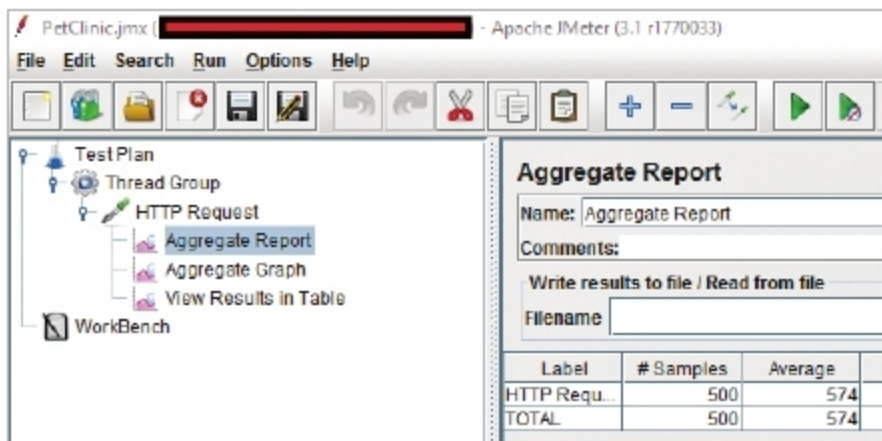


Figure 5: Apache JMeter HTTP request report

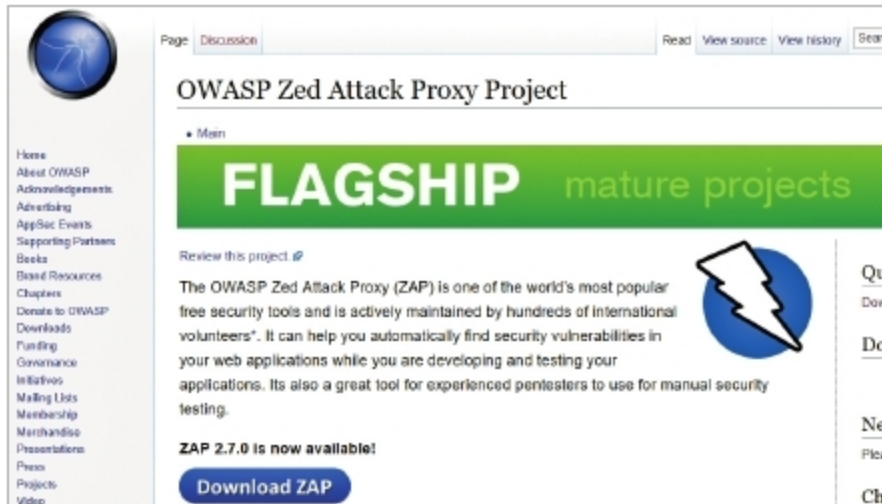


Figure 6: OWASP ZAP project

```
1.0.dtd">
```

```
<suite name="Suite">
<test name="Test">
<classes>
<class name="example.SampleTest"/>
</classes>
</test><!-- Test -->
</suite><!-- Suite -->
```

- Download the driver for a specific Web browser.
- Get the driver instance `driver.get` (`http://localhost:8090/petclinic/`).
- Get the *Title* of the page `String title = driver.getTitle();`
- We need to compare the selected test input values with the outcome of the functional test that we've executed.
 - `Assert.assertTrue(title.contains("Second Edition: Setting the stage for a DevOps culture"))`

The main advantage of functional test automation is that we can integrate it in the DevOps pipeline to avoid any manual intervention. Hundred per cent test coverage may not be feasible at present, but even 50 per cent automated functional tests can save a lot of effort and time. Additionally, over time, the team's confidence in automated functional tests will go up, which can change the culture of an organisation. Teams may soon go beyond 50 per cent coverage as DevOps related activities mature over time.

Automated functional tests play a vital role in shortening the time to market in DevOps (continuous testing). Such testing also makes a huge difference to the quality of

ZAP 2.7.0 Standard

Windows (64) Installer	2017-11-28	111 MB	Download now
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This contains the core set of functionality, and you can add more functionality at an Marketplace.

The Windows and Linux versions require Java 8 or higher to run.

Figure 7: Download ZAP based on OS

products, over time.

Load testing

Load testing is all about verifying the capacity of an application to serve users' demands in a specific interval of time. In load testing, typically, the usage of a Web application is measured by simulating multiple users. The application is accessed concurrently by creating threads.

Apache JMeter is an open source project and a pure Java application. It is used for Web application load tests. It helps to analyse and measure performance.

Name	Apache JMeter
Objective	Load testing tool to analyse and measure the performance of Web applications
Licence	Apache Licence 2.0
Stable release	5.0
Website	https://jmeter.apache.org/
Supported operating systems	Windows, Linux, Mac
Written in	Java
Support for integration with continuous integration tools	Commands can be easily integrated with CI tools such as Jenkins

Download Apache JMeter from http://jmeter.apache.org/download_jmeter.cgi.

Extract the files and go to the *bin* directory. Execute `jmeter.bat` or `jmeter.sh`.

- Open the Apache JMeter console. Create a *test plan*. Right-click on the *test plan* and click on *Add*. Select *Threads (Users)*.